

CONTACT

Paris, France
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EXPERTISE

Robotic perception

semantic SLAM, dynamic scenes, 3D LiDAR, tracking, map consistency

Embedded AI

TensorRT, CUDA-aware deployment, FP16 inference, embedded Linux, Jetson evaluation

Efficient deep learning

semantic and panoptic segmentation, model compression, latency and memory budgets

Engineering practice

C++ and Python robotics stacks, benchmarking, ROS, Docker, Git

STACK

C++ Python CUDA TensorRT
PyTorch OpenCV ROS Jetson Linux
Docker Git 3D LiDAR

EDUCATION

PhD, AI and Robotics 2023 – Present
CEA LIST / ENSTA Paris
Semantic Aware Dynamic SLAM for Embedded Systems

M2 MVA 2022 – 2023
ENS Paris-Saclay
Mathematics, Vision and Learning

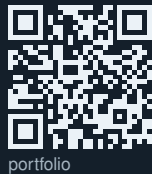
Engineering Degree 2020 – 2023
ESIEE Paris
AI, data science, electronics and embedded systems

LANGUAGES

French native
English professional
German intermediate

HOBBIES

Sports: equestrian (15 years, French nationals), boxing (3 years), running (marathons), swimming (triathlon)
Classical piano (14 years)
Chess
Entomology and myrmecology



portfolio

Calvin Galagain

Researcher & Engineer in Robotics, AI & Embedded Perception

PhD Candidate: Semantic Aware Dynamic SLAM for Embedded Systems

Robotics PhD researcher at CEA LIST and ENSTA Paris, focused on real-time semantic SLAM, dynamic-scene perception and efficient neural networks for resource-constrained robotic systems. I work at the boundary between research and engineering: designing perception algorithms, optimizing them for embedded deployment, and integrating them into C++/Python robotics stacks.

Research	Engineering	Embedded AI
semantic SLAM, dynamic scenes, efficient segmentation	C++/Python codebases, ROS, LiDAR tracking, benchmarking	TensorRT, CUDA, FP16, runtime and memory trade-offs

Research Experience

CEA LIST / ENSTA Paris 10/2023 – Present
Palaiseau, France

PhD Researcher in Embedded Robotic Perception

- Developing semantic SLAM methods for dynamic scenes, using semantic and panoptic segmentation to reduce moving-object artifacts and improve map consistency.
- Designing lightweight segmentation pipelines for robotics, with emphasis on predictable latency, memory footprint and deployability on constrained embedded platforms.
- Optimizing neural perception models for edge execution with TensorRT, CUDA-aware deployment, FP16 inference and NVIDIA Jetson evaluation.
- Benchmarking SLAM and segmentation trade-offs across accuracy, runtime and compute budget to guide embedded robotics architecture decisions.

Exwayz 04/2023 – 10/2023
Paris, France

AI Research Intern, 3D Robotic Perception

- Built real-time object detection and tracking components from 3D LiDAR data for robotic perception stacks.
- Worked on SLAM, SLAM-GPS fusion, anisotropic point-cloud processing and dynamic bounding-box estimation in production-oriented Python/C++ codebases.

Zentrum für Recht und Digitalisierung 05/2022 – 09/2022
Saarbrücken, Germany

AI Research Intern

- Built real-time video-analysis tools for sensitive-content detection in collaboration with legal experts.
- Explored adversarial learning techniques to assess vulnerabilities in deep neural networks.

IRCGN 11/2021 – 09/2022
Pontoise, France

Computer Vision Research Project

- Studied deepfake detection and forensic face recognition from police video footage.
- Implemented landmark detection, data augmentation and deep-learning decoders from scratch.

Selected Publications

Token-Space Mask Prediction for Efficient Vision Transformer Segmentation	CVPR, 2026
Calvin Galagain, Martyna Poreba, François Goulette	
LiPS: Lightweight Panoptic Segmentation for Resource-Constrained Robotics	IEEE ICIP, 2026
Calvin Galagain, Martyna Poreba, François Goulette, Cyrill Stachniss	
Is Semantic SLAM Ready for Embedded Systems? A Comparative Survey	IEEE RAS, 2025
Calvin Galagain, Martyna Poreba, François Goulette	
Challenges and Performance of SLAM Algorithms on Resource-Constrained Devices	EEAI, 2025
Calvin Galagain, Martyna Poreba, François Goulette	
Optimizing Vision Transformers for Edge Deployment	EEAI, 2025
Mathilde Proust, Martyna Poreba, Calvin Galagain, Michal Szczepanski, Karim Haroun	

Teaching

GPGPU Programming for ML	3D Point Clouds
ENSTA Paris / Telecom Paris	MVA
AI and Data	C Programming
ENSTA Paris	ENSTA Paris